SAULO QUIÑONES GÓNGORA

Applied AI | GIS | Web Development

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Portfolio • GitHub • LinkedIn

PROFESSIONAL PROFILE

Engineer working across three areas, applied artificial intelligence, geographic information systems (GIS) and web development. Trained as an Environmental Engineer, with a specialization in Artificial Intelligence and an IBM professional certificate in Data Science. Builds data pipelines, spatial databases and AI-enabled applications in which every result can be traced and rechecked, from raw, inconsistent data and backend logic to a deployed interface.

EXPERIENCE

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE (UAO) WITH ASOCAÑA — Cali, Colombia
GIS Analyst and Environmental Engineer, Biological Corridor Project — Feb 2026 – Present

- Integrated environmental data for the Cauca River and its 15 priority tributaries from four authorities (CVC, CARDER, CRC, IDEAM), each with different formats, coordinate systems, and station names, using Python scripts (pandas, pyproj) that automate cleaning, schema harmonization, and station matching.
- Built the project's central geodatabase in ArcGIS Pro (water quality, streamflow, land use, and monitoring stations) under a single coordinate reference system, as the single source for all downstream analysis.
- Implemented reproducible data quality controls: filters for physically implausible values, detection of duplicate records across stations, and a log documenting every exclusion.
- Analyzed historical streamflow and water quality series (2015–2026 campaigns): flow duration curves, hydrological classification by exceedance percentiles, and longitudinal profiles of six physicochemical parameters, all generated with scripts.
- Modeled diffuse nitrogen and phosphorus load across 46 sub-reaches under three scenarios in a formula-linked Excel model with sensitivity analysis; 7 sub-reaches (15%) concentrate 60% of the nitrogen load.
- Developed Cauca River Baseline, a public viewer built with MapLibre GL JS (JavaScript) that exposes monitoring stations, water quality profiles, and flow duration curves, with CSV download per station, to share results with institutional and industry partners.

FUDAMBIENT, CONTRACTOR FOR CVC (Regional Environmental Authority) — Cali, Colombia
Environmental Professional, CVC Agreements N°031 and N°094 — Sep 2025 – Jan 2026

- Provided technical and pedagogical support under CVC Agreement N°031 to strengthen rural wastewater treatment systems (SITAR) as nature-based solutions, working directly with communities on implementation and long-term sustainability.
- Automated internal project workflows with Google Apps Script and built interactive web tools for environmental education, including an interactive map and the institutional website (HTML, CSS, JavaScript), the last month as freelance work.

ITESA, INSTITUTO TECNOLÓGICO SUPERIOR DE HIDALGO — Hidalgo, Mexico (Remote)
Remote Research Collaborator, Programa Delfín 2025 — Jun 2025 – Jul 2025

- Designed and built OptiCoder, a React Native and Expo mobile accessibility app integrating multimodal AI (Google Gemini for vision-language, OpenAI Whisper for speech-to-text) to help visually impaired developers interpret code screenshots, UI mockups, and architecture diagrams, with a layered architecture and a two-stage prompting strategy.
- Designed and ran an empirical evaluation with 7 software developers across 106 classification instances, reaching 74.1% precision and 58.1% recall in code-error detection, and revised the prompt design directly from the findings, including a fix for the model misreading IDE error highlights instead of the code itself.
- Co-authored the resulting research paper, currently under review at an indexed journal in AI and accessibility.

ASOASIDE, CONTRACTOR FOR CVC — Cali, Colombia
Environmental Professional, CVC Agreement N°004 — Mar 2025 – Sep 2025

- Supported socio-environmental diagnostics of river basins across the Valle del Cauca using participatory methods (social cartography, problem trees) to prioritize needs by municipality.
- Authored a school environmental education plan (PRAE) and produced technical documentation and tracking matrices for municipal environmental education plans (PMEA), following CVC guidelines; automated project workflows with Google Apps Script.

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE (UAO) — Cali, Colombia
Undergraduate Researcher, Semillero A+ — Jan 2024 – Nov 2024

- Built statistical and data mining models to assess the hydrobiological water quality of the Cali River for the undergraduate thesis, using biotic indices (BMWP/Col), habitat suitability models, and PCA in Python and XLSTAT.

- Analyzed raw air quality monitoring data (PM_{10} and $PM_{2.5}$) from Cali's MIO public transit system, including exploratory and statistical analysis and mapping in Python (pandas, Matplotlib, Folium); presented the analysis at RedCOLSI, scoring 94/100.

SELECTED PROJECTS

- **RiverEtch** (Submitted to the IEEE ClimateChain Global Hackathon, Oct 2026). Full-stack system that estimates river turbidity and water temperature from Sentinel-2 and Landsat imagery, with stated uncertainty, and seals each reading in a hash-chained ledger anchored on an Ethereum testnet and timestamped on Bitcoin, so any later edit is detectable. Frozen models scored once on held-out years against USGS gauges (median error of 18% and 20% at the two best rivers, 33% across the nine evaluated), with every validation result recomputed in CI. Python, TypeScript, React, Solidity, MapLibre GL JS. https://github.com/saulo1112/RiverEtch
- **OptiCoder.** Multimodal AI mobile app that reads code screenshots, UI mockups and diagrams aloud for developers who cannot see them, evaluated with 7 developers. React Native, Expo, Gemini, Whisper. https://github.com/saulo1112/OptiCoder
- **Cloud and IoT deployments on AWS.** Team projects from the AI specialization. A serverless IoT pipeline (ESP32 sensors, IoT Core, DynamoDB, Lambda, API Gateway, S3), a sentiment analysis app across two availability zones behind a load balancer, and a three-service image classifier on EKS. https://github.com/saulo1112/sentiments-app
- **EUDR Forest Risk Assessment.** Screens 4,170 real agricultural parcels for EU deforestation-free compliance from open satellite data, with results stored in PostGIS and served through a REST API. Python, PostGIS, Google Earth Engine, scikit-learn, FastAPI, Docker. https://github.com/saulo1112/EUDR_Risk_Assessment
- **App Physical Recovery.** Team project of four, a webcam physiotherapy coach built on person detection and pose estimation. Contributed the frame preprocessing module, tests, README, docstrings and Makefile, and edited the final report. PyTorch, HuggingFace, gRPC, Streamlit, Docker. https://github.com/saulo1112/App_PhysicalRecovery

EDUCATION AND CERTIFICATIONS

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE Cali, Colombia
Specialization in Artificial Intelligence Completed Oct 2026

- GPA 4.9/5.0 (Colombian 0–5 scale). Coursework in machine learning, deep learning, computer vision, time-series forecasting, and applied AI systems.

B.Eng. in Environmental Engineering Graduated May 2025

- GPA 4.7/5.0. Academic Excellence Award in semesters 2021-1, 2022-1, 2024-1, and 2024-2.

High school valedictorian, Colegio Los Andes

CERTIFICATIONS

- IBM Data Science Professional Certificate: Python, SQL, data analysis, and machine learning (https://coursera.org/verify/professional-cert/QWUM9USDXC33).
- EF SET English Certificate, level C2 (https://cert.efset.org/en/7jVogB).
- AWS Certified Machine Learning Engineer, Associate (in progress, exam scheduled for December 2026).

SKILLS

AI and machine learning: PyTorch, scikit-learn, OpenCV, multimodal models (Gemini, Whisper), prompt evaluation, Claude Code.
Programming and databases: Python (pandas, NumPy, GeoPandas, rasterio), SQL, PostgreSQL/PostGIS, JavaScript, TypeScript, R.
Geospatial and remote sensing: Sentinel-2, Landsat, Google Earth Engine, STAC, ArcGIS Pro (arcpy), QGIS, GDAL, pyproj.
Cloud and deployment: AWS (EC2, VPC, EKS, Lambda, IoT Core, DynamoDB, API Gateway, S3), Docker, GitHub Actions, pytest.
APIs, web and visualization: FastAPI, Flask, Node.js, React, Vite, React Native, MapLibre GL JS, Matplotlib, Folium, Power BI.
Other tools: Solidity (Hardhat, ethers.js), Vercel, XLSTAT, Qual2k, Google Apps Script, advanced spreadsheet automation.
Languages: Spanish (native), English (C2).

VOLUNTEERING

UN VOLUNTEERS COLOMBIA Cali, Colombia
Logistics Assistant, COP16 Cali Oct 2024 – Nov 2024

- Coordinated participant guidance and transport logistics for a large-scale international summit, communicating bilingually in Spanish and English with delegations from multiple countries.